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# Generalized Least Squares (GLS): Relations to OLS & WLS

with Mathematical Derivations & Intuition



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116



2



## Background and Motivation

For anyone pursuing study in Statistics or Machine Learning, Ordinary Least Squares (OLS) Linear Regression is one of the first and most “simple” methods one is exposed to. However, there are a set of mathematical restrictions under which the OLS estimator is the Best Linear Unbiased Estimator (BLUE), i.e. the unbiased estimator with minimal sampling variance. (*For a more thorough overview of OLS, the BLUE, and the Gauss-Markov Theorem, please see my previous piece on the subject*)

What if the mathematical assumptions for the OLS being the BLUE do not hold? The next “leap” is **Generalized Least Squares (GLS)**, of which the OLS is in fact a special case of. This article serves as an introduction to GLS, with the following topics covered:

1. Review of the OLS estimator and conditions required for it to be BLUE
2. Mathematical set-up for Generalized Least Squares (GLS)
3. Recovering the GLS estimator
4. Proof that the GLS Estimator is Unbiased
5. Recovering the variance of the GLS estimator
6. Short discussion on relation to Weighted Least Squares (WLS)

Note, that in this article I am working from a Frequentist paradigm (as opposed to a Bayesian paradigm), mostly as a matter of convenience. So, let’s jump in:

## 1. Review of the OLS estimator and conditions for it being the BLUE

## Let's start with a quick review of the OLS estimator

Let's say we have the following additive linear model:

$$\underline{Y} = \underline{X}\underline{\beta} + \underline{\epsilon} \quad , \quad \text{with} \quad \underline{Y} \in \mathbb{R}^{n \times 1} ; \underline{\beta} \in \mathbb{R}^{k \times 1} ; \underline{X} \in \mathbb{R}^{n \times k} ; \underline{\epsilon} \in \mathbb{R}^{n \times 1}$$

Where:

- $Y_i$  is random variable
- $\beta_j$  is a non-random but unobservable parameter
- $X_{ij}$  is a non-random observable parameter (called an “explanatory variable”)
- $\epsilon_i$  is a random variable and is the “random error”, i.e.  $\epsilon_i = Y_i - \sum_{j=1}^k X_{ij}\beta_j$

And the standard Gauss-Markov Assumptions hold:

1.  $E[\epsilon_i] = 0$
2.  $Var(\epsilon_i) = \sigma^2 < \infty, \forall i$
3.  $Cov(\epsilon_i, \epsilon_j) = 0, \forall i \neq j$

Descriptively, the assumptions above state that all  $n$  error terms  $\epsilon_i$  are uncorrelated, have the same variance, and have expectation zero.

Suppose we now have some data, and we would like to “fit our data” to the additive linear model we have specified. Now we have:

$$\underline{Y} = \underline{X}\underline{\hat{\beta}} + \underline{e} \quad \text{with} \quad \underline{Y} \in \mathbb{R}^{n \times 1} ; \underline{\hat{\beta}} \in \mathbb{R}^{k \times 1} ; \underline{X} \in \mathbb{R}^{n \times k} ; \underline{e} \in \mathbb{R}^{n \times 1}$$

$$\underline{\hat{Y}} = \underline{X}\underline{\hat{\beta}} \quad \text{with} \quad \underline{\hat{Y}} \in \mathbb{R}^{n \times 1} ; \underline{\hat{\beta}} \in \mathbb{R}^{k \times 1} ; \underline{X} \in \mathbb{R}^{n \times k}$$

In the notation above:

- $\underline{\hat{\beta}}$  is a vector of “candidate value” or “estimates” for the parameter vector  $\underline{\beta}$ .
- The quantity  $\underline{Y} - \underline{X}\underline{\hat{\beta}}$  is the residual vector  $\underline{e}$ , and for each observation measures the vertical distance between the actual datapoint  $Y_i$  and it's prediction  $\hat{Y}_i$  which lies on the hyperplane.

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To estimate the “best candidate” vector  $\underline{\hat{\beta}}$ , we can minimize the sum of squared residuals (SSR) with respect to  $\underline{\hat{\beta}}$ . Doing so will give us the OLS estimator:

$$\begin{aligned} SSR &= \underline{e}^T \underline{e} \\ &= (\underline{Y} - \underline{X}\underline{\hat{\beta}})^T (\underline{Y} - \underline{X}\underline{\hat{\beta}}) \\ &= (\underline{Y}^T - \underline{\hat{\beta}}^T \underline{X}^T) (\underline{Y} - \underline{X}\underline{\hat{\beta}}) \\ &= \underline{Y}^T \underline{Y} - \underline{Y}^T \underline{X}\underline{\hat{\beta}} - \underline{\hat{\beta}}^T \underline{X}^T \underline{Y} + \underline{\hat{\beta}}^T \underline{X}^T \underline{X}\underline{\hat{\beta}} \\ &= \underline{Y}^T \underline{Y} - \underline{\hat{\beta}}^T \underline{X}^T \underline{Y} - \underline{\hat{\beta}}^T \underline{X}^T \underline{Y} + \underline{\hat{\beta}}^T \underline{X}^T \underline{X}\underline{\hat{\beta}} \\ &= \underline{Y}^T \underline{Y} - 2\underline{\hat{\beta}}^T \underline{X}^T \underline{Y} + \underline{\hat{\beta}}^T \underline{X}^T \underline{X}\underline{\hat{\beta}} \end{aligned}$$

Note that  $\underline{\hat{\beta}} = \text{argmin}_{\underline{\hat{\beta}}} SSR$ :

$$\begin{aligned} \frac{\partial(SSR)}{\partial \underline{\hat{\beta}}} &= \frac{\partial}{\partial \underline{\hat{\beta}}} (\underline{Y}^T \underline{Y} - 2\underline{\hat{\beta}}^T \underline{X}^T \underline{Y} + \underline{\hat{\beta}}^T \underline{X}^T \underline{X}\underline{\hat{\beta}}) \\ &= -2\underline{X}^T \underline{Y} + 2\underline{X}^T \underline{X}\underline{\hat{\beta}} \xrightarrow{set} 0 \end{aligned}$$

$$\underline{X}^T \underline{X}\underline{\hat{\beta}} = \underline{X}^T \underline{Y}$$

$$\underline{\hat{\beta}} = (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{Y}$$

This gives us the OLS estimator:

$$\underline{\hat{\beta}}_{OLS} = (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{Y}$$

The predicted values  $\underline{\hat{Y}}$  are therefore:  $\underline{\hat{Y}} = \underline{X}\underline{\hat{\beta}}_{OLS} = \underline{X}(\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{Y} = \underline{P}\underline{Y}$

The matrix  $\underline{P} = \underline{X}(\underline{X}^T \underline{X})^{-1} \underline{X}^T$  is called the “projection matrix” or the “hat matrix” (because it’s “puts the hat  $\hat{\cdot}$  on  $\underline{Y}$ ”).

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For further information on the OLS estimator and proof that it’s unbiased, please see [my previous piece on the subject](#).

## 2. Mathematical set-up for Generalized Least Squares (GLS)

Suppose we have:

$$\underline{Y} = \underline{X}\underline{\beta} + \underline{\epsilon} \quad \text{with} \quad \underline{Y}, \underline{\epsilon} \in \mathbb{R}^{1 \times n}; \underline{\beta} \in \mathbb{R}^{1 \times k}; \underline{X} \in \mathbb{R}^{n \times k}$$

$$\underline{Y} = \underline{X}\underline{\hat{\beta}} + \underline{e} \quad \text{with} \quad \underline{Y}, \underline{e} \in \mathbb{R}^{1 \times n}; \underline{\hat{\beta}} \in \mathbb{R}^{1 \times k}; \underline{X} \in \mathbb{R}^{n \times k}$$

$$\underline{\hat{Y}} = \underline{X}\underline{\hat{\beta}} \quad \text{with} \quad \underline{\hat{Y}}, \underline{e} \in \mathbb{R}^{1 \times n}; \underline{\hat{\beta}} \in \mathbb{R}^{1 \times k}; \underline{X} \in \mathbb{R}^{n \times k}$$

With  $E[\underline{\epsilon}] = 0$ , and  $Var(\underline{\epsilon}) = \sigma^2 \underline{V}$ , where  $\underline{V}$  is a known  $n \times n$  matrix. If  $\underline{V}$  is diagonal but with unequal elements, the observations in  $\underline{Y}$  are uncorrelated but have unequal variance, while if  $\underline{V}$  has non-zero off-diagonal elements, at least some of the observations in  $\underline{y}$  are correlated.

Let's say that matrix  $\underline{V}$  is either non-diagonal, or is diagonal but with diagonal elements unequal. In this case the OLS estimate  $\underline{\hat{\beta}}_{OLS} = (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{Y}$ , although still unbiased, is not statistically efficient (it does not have minimum sampling variance and is not the BLUE). Therefore, to recover the BLUE estimator an approach is to transform the model to a new set of observations that satisfy the constant variance assumption, and then use OLS with these transformed parameters. We call the resulting estimator the **Generalized Least Squares (GLS) Estimator**.

Given that  $\sigma^2 \underline{V}$  is a covariance matrix, we know that  $\underline{V}$  must be symmetric and non-singular. Therefore we can define:

$$\underline{V} = \underline{K}^T \underline{K} = \underline{K} \underline{K}$$

with  $\underline{K}$  called the "square root" of  $\underline{V}$

Let us define:

- $\underline{Y}' = \underline{K}^{-1} \underline{Y}$
- $\underline{X}' = \underline{K}^{-1} \underline{X}$
- $\underline{\epsilon}' = \underline{K}^{-1} \underline{\epsilon}$
- $\underline{e}' = \underline{K}^{-1} \underline{e}$

Note, that:

- $E[\underline{\epsilon}'] = E[\underline{K}^{-1} \underline{\epsilon}] = \underline{K}^{-1} E[\underline{\epsilon}] = \underline{K}^{-1} E[\underline{\epsilon}] = \underline{K}^{-1} (0) = 0$
- $Var(\underline{\epsilon}') = Var(\underline{K}^{-1} \underline{\epsilon}) = \underline{K}^{-1} Var(\underline{\epsilon}) \underline{K}^{-1} = \underline{K}^{-1} \sigma^2 \underline{V} \underline{K}^{-1} = \sigma^2 \underline{K}^{-1} \underline{K} \underline{K}^{-1} = \sigma^2 \underline{I}$

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### 3. Recovering the GLS estimator



To estimate the “best candidate”  $\hat{\underline{\beta}}$ , we can minimize the sum of squared “adjusted” residuals (SSR) with respect to  $\underline{\hat{\beta}}$ :

$$\begin{aligned}
 SSR &= \underline{e}'^T \underline{e}' \\
 &= (\underline{Y}' - \underline{X}' \underline{\hat{\beta}})^T (\underline{Y}' - \underline{X}' \underline{\hat{\beta}}) \\
 &= (\underline{K}^{-1} \underline{Y} - \underline{K}^{-1} \underline{X} \underline{\hat{\beta}})^T (\underline{K}^{-1} \underline{Y} - \underline{K}^{-1} \underline{X} \underline{\hat{\beta}}) \\
 &= (\underline{K}^{-1} (\underline{Y} - \underline{X} \underline{\hat{\beta}}))^T (\underline{K}^{-1} (\underline{Y} - \underline{X} \underline{\hat{\beta}})) \\
 &= (\underline{Y} - \underline{X} \underline{\hat{\beta}})^T \underline{K}^{-1} \underline{K}^{-1} (\underline{Y} - \underline{X} \underline{\hat{\beta}}) \\
 &= (\underline{Y} - \underline{X} \underline{\hat{\beta}})^T \underline{V}^{-1} (\underline{Y} - \underline{X} \underline{\hat{\beta}}) \\
 &= (\underline{Y}^T - \underline{\hat{\beta}}^T \underline{X}^T) \underline{V}^{-1} (\underline{Y} - \underline{X} \underline{\hat{\beta}}) \\
 &= \underline{Y}^T \underline{V}^{-1} \underline{Y} - \underline{Y}^T \underline{V}^{-1} \underline{X} \underline{\hat{\beta}} - \underline{\hat{\beta}}^T \underline{X}^T \underline{V}^{-1} \underline{Y} + \underline{\hat{\beta}}^T \underline{X}^T \underline{V}^{-1} \underline{X} \underline{\hat{\beta}} \\
 &= \underline{Y}^T \underline{V}^{-1} \underline{Y} - \underline{\hat{\beta}}^T \underline{X}^T \underline{V}^{-1} \underline{Y} - \underline{\hat{\beta}}^T \underline{X}^T \underline{V}^{-1} \underline{Y} + \underline{\hat{\beta}}^T \underline{X}^T \underline{V}^{-1} \underline{X} \underline{\hat{\beta}} \\
 &= \underline{Y}^T \underline{V}^{-1} \underline{Y} - 2 \underline{\hat{\beta}}^T \underline{X}^T \underline{V}^{-1} \underline{Y} + \underline{\hat{\beta}}^T \underline{X}^T \underline{V}^{-1} \underline{X} \underline{\hat{\beta}}
 \end{aligned}$$

Note that  $\underline{\hat{\beta}} = \text{argmin}_{\underline{\hat{\beta}}} SSR$ :

$$\begin{aligned}
 \frac{\partial(SSR)}{\partial \underline{\hat{\beta}}} &= \frac{\partial}{\partial \underline{\hat{\beta}}} (\underline{Y}^T \underline{V}^{-1} \underline{Y} - 2 \underline{\hat{\beta}}^T \underline{X}^T \underline{V}^{-1} \underline{Y} + \underline{\hat{\beta}}^T \underline{X}^T \underline{V}^{-1} \underline{X} \underline{\hat{\beta}}) \\
 &= -2 \underline{X}^T \underline{V}^{-1} \underline{Y} + 2 \underline{X}^T \underline{V}^{-1} \underline{X} \underline{\hat{\beta}} \stackrel{set}{\rightarrow} 0
 \end{aligned}$$

$$\underline{X}^T \underline{V}^{-1} \underline{X} \underline{\hat{\beta}} = \underline{X}^T \underline{V}^{-1} \underline{Y}$$

$$\underline{\hat{\beta}} = (\underline{X}^T \underline{V}^{-1} \underline{X})^{-1} \underline{X}^T \underline{V}^{-1} \underline{Y}$$

This gives us the GLS estimator:

$$\underline{\hat{\beta}}_{GLS} = (\underline{X}^T \underline{V}^{-1} \underline{X})^{-1} \underline{X}^T \underline{V}^{-1} \underline{Y}$$

The predicted values  $\hat{\underline{Y}}$  are therefore:  $\hat{\underline{Y}} = \underline{X} \underline{\hat{\beta}} = \underline{X} (\underline{X}^T \underline{V}^{-1} \underline{X})^{-1} \underline{X}^T \underline{V}^{-1} \underline{Y} = \underline{P} \underline{Y}$

In this case,  $\underline{P} = \underline{X} (\underline{X}^T \underline{V}^{-1} \underline{X})^{-1} \underline{X}^T \underline{V}^{-1}$  is the “projection matrix” or “hat matrix”

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## 4. Proof that the GLS Estimator is Unbiased

Show that  $\underline{\hat{\beta}}_{GLS}$  is an unbiased estimate of  $\underline{\beta}$ :

$$\begin{aligned}
 E[\underline{\hat{\beta}}_{GLS}] &= E[(\underline{X}^T \underline{V}^{-1} \underline{X})^{-1} \underline{X}^T \underline{V}^{-1} \underline{Y}] \\
 &= E[(\underline{X}^T \underline{V}^{-1} \underline{X})^{-1} \underline{X}^T \underline{V}^{-1} (\underline{X} \underline{\beta} + \underline{\epsilon})] \\
 &= E[(\underline{X}^T \underline{V}^{-1} \underline{X})^{-1} \underline{X}^T \underline{V}^{-1} \underline{X} \underline{\beta} + (\underline{X}^T \underline{V}^{-1} \underline{X})^{-1} \underline{X}^T \underline{V}^{-1} \underline{\epsilon}] \\
 &= E[\underline{\beta} + (\underline{X}^T \underline{V}^{-1} \underline{X})^{-1} \underline{X}^T \underline{V}^{-1} \underline{\epsilon}] \\
 &= E[\underline{\beta}] + ((\underline{X}^T \underline{V}^{-1} \underline{X})^{-1} \underline{X}^T \underline{V}^{-1}) E[\underline{\epsilon}] \\
 &= \underline{\beta} + ((\underline{X}^T \underline{V}^{-1} \underline{X})^{-1} \underline{X}^T \underline{V}^{-1})(0) \\
 &= \underline{\beta}
 \end{aligned}$$

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## 5. Recovering the variance of the GLS estimator

Recover variance of  $\hat{\beta}_{GLS}$ :

Note that:

$$\begin{aligned}\hat{\beta}_{GLS} &= (X^T V^{-1} X)^{-1} X^T V^{-1} y \\ &= (X^T V^{-1} X)^{-1} X^T V^{-1} (X\beta + \epsilon) \\ &= \beta + (X^T V^{-1} X)^{-1} X^T V^{-1} \epsilon \\ \hat{\beta}_{GLS} - \beta &= (X^T V^{-1} X)^{-1} X^T V^{-1} \epsilon\end{aligned}$$

$$\begin{aligned}Var(\hat{\beta}_{GLS}) &= E\left[\left(\hat{\beta} - E[\hat{\beta}]\right)^2\right] \\ &= E\left[\left(\hat{\beta} - \beta\right)^2\right] \\ &= E\left[\left(\hat{\beta} - \beta\right)\left(\hat{\beta} - \beta\right)^T\right] \\ &= E\left[\left((X^T V^{-1} X)^{-1} X^T V^{-1} \epsilon\right)\left((X^T V^{-1} X)^{-1} X^T V^{-1} \epsilon\right)^T\right] \\ &= E\left[\left((X^T V^{-1} X)^{-1} X^T V^{-1} \epsilon\right)\left(\epsilon^T V^{-1} X (X^T V^{-1} X)^{-1}\right)\right] \\ &= E\left[\left(X^T V^{-1} X\right)^{-1} X^T V^{-1} \epsilon \epsilon^T V^{-1} X (X^T V^{-1} X)^{-1}\right] \\ &= (X^T V^{-1} X)^{-1} X^T V^{-1} E[\epsilon \epsilon^T] V^{-1} X (X^T V^{-1} X)^{-1} \\ &= (X^T V^{-1} X)^{-1} X^T V^{-1} Var(\epsilon) V^{-1} X (X^T V^{-1} X)^{-1} \\ &= (X^T V^{-1} X)^{-1} X^T V^{-1} \sigma^2 V V^{-1} X (X^T V^{-1} X)^{-1} \\ &= \sigma^2 (X^T V^{-1} X)^{-1} X^T V^{-1} V V^{-1} X (X^T V^{-1} X)^{-1} \\ &= \sigma^2 (X^T V^{-1} X)^{-1} X^T V^{-1} X (X^T V^{-1} X)^{-1} \\ &= \sigma^2 (X^T V^{-1} X)^{-1}\end{aligned}$$

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## 6. Short discussion on relation to Weighted Least Squares (WLS)

Weighted Least Squares (WLS):

One of the drawbacks of GLS is that we must specify the matrix  $V$ . This matrix is typically not known, and specification can be difficult.

WLS is a special case of GLS, and makes some simplifying assumptions about matrix  $V$ , making it's specification a bit more reasonable. With WLS we assume that the errors  $\epsilon$  are uncorrelated, but do not have equal variances. Therefore with  $Var(\epsilon) = \sigma^2 V$ , matrix  $V$  is a strictly diagonal matrix with entries proportional to the variances of observations in  $y$ . Observations with smaller variances will receive larger weights in  $V$ , with observations with larger variances receiving smaller weights. One example of possible weights include  $w_i \propto e_i$ . Therefore, observations with larger residuals will receive smaller weights in the matrix.

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## Wrap-up, Final Thoughts, and Next Steps

Generalized Least Squares (GLS) is a large topic. This article serves as a short introduction meant to “set the scene” for GLS mathematically. There's plenty more to be covered, including (but not limited to):

1. A more in-depth dive into WLS
2. Methods and approaches for specifying covariance matrix  $V$
3. The topic of Feasible Generalized Least Squares
4. Relation to Iteratively Reweighted Least Squares (IRLS)

I plan on covering these topics in-depth in future pieces.

I hope the above is insightful. As I've mentioned in some of my previous pieces, it's my opinion not enough folks take the time to go through these

types of exercises. For me, this type of theory-based insight leaves me more comfortable using methods in practice. A personal goal of mine is to encourage others in the field to take a similar approach. I’m planning on writing based pieces in the future, so feel free to connect with me on [LinkedIn](#), and [follow me here on Medium](#) for updates!

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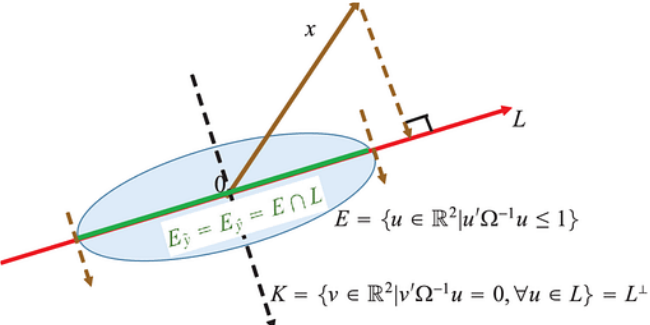
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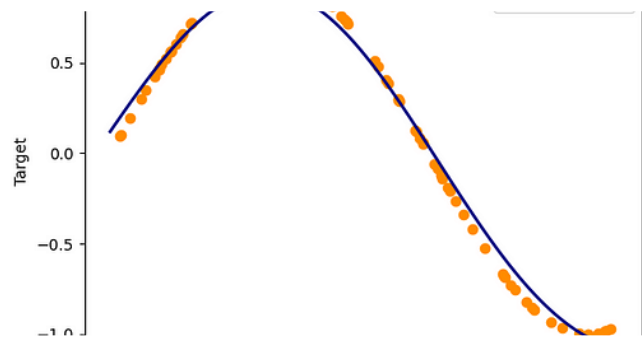




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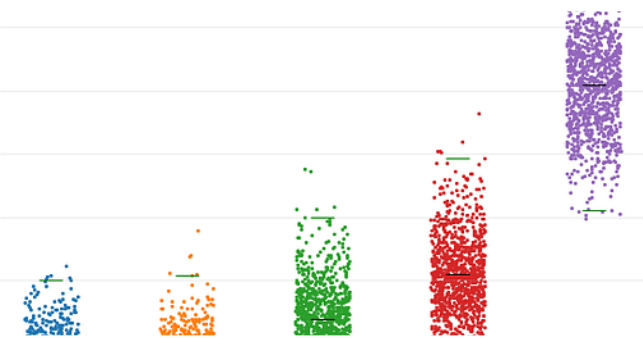
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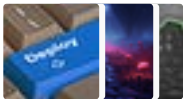
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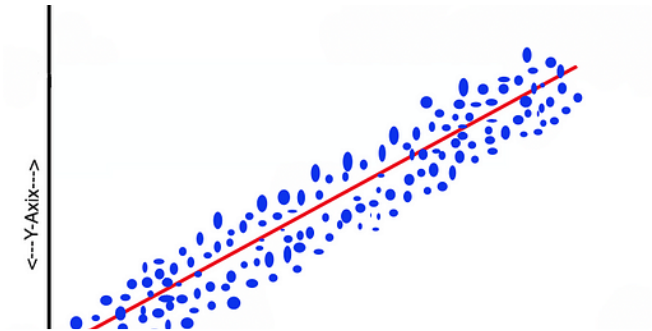
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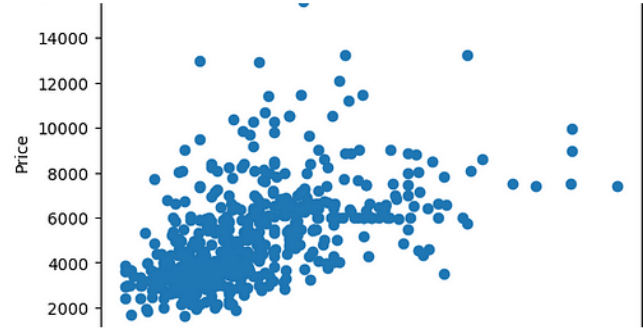
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